

Alpha decay 2

Let our hypothesis be that the nucleus produces alpha particles at an average rate ν which then tunnel with probability T . Hence the decay rate λ is

$$\lambda = \nu T$$

The half-life $\tau_{1/2}$ is

$$\tau_{1/2} = \frac{\log 2}{\lambda} = \frac{\log 2}{\nu T}$$

We seek to calculate $\tau_{1/2}$ from T . The problem is that ν is only known empirically so we use the Geiger-Nuttall formula

$$\log_{10} \tau_{1/2} = -C_1 - C_2 Z^{2/3} + C_3 \frac{Z}{\sqrt{E}} \quad (1)$$

where

$$C_1 = 28.9$$

$$C_2 = 1.60$$

$$C_3 = 1.61$$

For the alpha decay process ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th}$ we have

$$Z = 90$$

$$E = 4.27 \text{ MeV}$$

Hence

$$\tau_{1/2} = 1.2 \times 10^9 \text{ yr}$$

The actual half-life of ${}^{238}\text{U}$ is

$$4.463 \times 10^9 \text{ yr}$$

Eigenmath script